

Name of Student: Darshan JainScholar No.: 290030608**INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL**

Department of Information Technology

End-Term Examination November – 2025

Branch:- B. Tech IT

Semester –III

Subject Name/ Code: - Computer Organization and Architecture / IT-3005

Max. Marks: 60

Time Allotted: 180 min

NOTE: 1. All questions are compulsory.
2. Mark answers number clearly.

S. No	Question	Marks
1.	(a) Distinguish MAR and MDR ; explain briefly why both are needed in memory access.	3
	(b) Trace all microoperations involved in executing this instruction: MOV R3, [4500] Name all registers used and explain the flow.	4
	(c) If both operands of an instruction are in memory instead of registers, analyse how execution time changes with respect to memory accesses.	3
2.	(a) Describe how the Sequence Counter and timing pulses (T_0 , T_1 , T_2 ...) coordinate instruction steps.	3
	(b) Given this microoperation flow: T_0 : $AR \leftarrow PC$ T_1 : $IR \leftarrow M[AR]$ T_2 : $PC \leftarrow PC + 1$ Write the corresponding control signals for each step.	3
	(c) Derive the functional meaning of the control equation: $LD(MAR) = R'T_0 + R'T_2 + D_7 I T_3$ Explain what each term represents in control sequencing.	4
3.	(a) Compare microprogrammed and hardwired control by identifying one key advantage and one limitation of the microprogrammed approach. How does each affect system flexibility or speed?	2
	(b) Explain the significance of the M CF BA fields in a microinstruction. How do these fields influence next-address sequencing?	3
	(c) Outline a simple conceptual flow of how a microprogrammed control unit performs the fetch phase and then returns to it after executing an instruction. Highlight design choices that ensure smooth sequencing.	5

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4.	(a)	Clearly distinguish between temporal locality and spatial locality. Give one simple example for each to show how they work in real programs.	3
	(b)	A cache has: • 32-bit address • Cache size = 8 KB • Block size = 32 bytes Compute precisely: - Number of cache blocks. - Number of index bits. - Number of offset bits. - Number of tag bits. Show all working (binary arithmetic) and explain how you derived each field size.	4
	(c)	Explain why TLB significantly improves virtual memory performance and analyse what happens if the TLB hit rate is extremely low.	3
5.	(a)	Explain the term load-store architecture and give one machine instruction (in assembly-like syntax) that demonstrates it. Explain why the instruction is classified as load-store.	3
	(b)	A task needs: • 1,000,000 instructions on CISC, CPI = 4 • 1,400,000 instructions on RISC, CPI = 1.2 • Clock = 2 GHz Compute: - Execution time on each - Which is faster and by what factor?	4
	(c)	Modern Intel processors internally convert CISC instructions into RISC-like micro-ops. Explain why this technique is used and evaluate whether this makes Intel processors "RISC".	3
6.	(a)	What happens during a subroutine call? Explain how the stack and program counter work together to make the call possible.	2
	(b)	A processor uses a stack-based subroutine call mechanism. Given the following nested calls: • CALL SUB1 at address 300 • Inside SUB1: CALL SUB2	3
	(c)	Compare maskable and non-maskable interrupts in terms of their priority and effect on program execution. Provide one scenario where improper interrupt handling might corrupt subroutine flow — consider what happens if the stack or return address is not properly saved before an interrupt occurs and explain why.	5